

Handling Very Large and Small Numbers

Synopsis: Physics is a quantitative science – we put numbers to things. For example, the speed of light is 300000000 m/s (8 zeros). The mass of the Earth is 5980000000000000000000 kg (22 zeros). The diameter of an atom is approximately 0.000000001 m (9 zeros after the decimal point). As you can see, we deal with extremely large to extremely small and everything in between. If you write the numbers as I have, it gets rather annoying. We must devise a system to represent the large and small numbers. The solution is to use "**powers of 10**" notation. For example, 300000000 m/s is 3×10^8 m/s. We pay attention only to "meaningful" numbers (all nonzero numbers and zeros sandwiched between nonzero numbers) and write the remaining zeros as the powers of 10. So, the mass of the Earth is 598×10^{22} kg, and the diameter of an atom is 1×10^{-10} m. Note that decimals have negative powers of 10. As you can see, this way of writing numbers is much more convenient and less cluttered. Combining the power of 10 notations with **scientific notation** (see below), we have the standard way of writing large and small numbers. In this note, we practice these notations and correctly use calculator buttons to input these large and small numbers (the "EE" button).

Performance Expectations: Write numbers in powers of 10 notation and scientific notation. Recognize significant numbers. Use the EE button on the calculator to perform calculations involving very large and small numbers.

Focus on Science Practices: Using mathematics and computational thinking, Scale, proportion, and models.

Instructions and Questions

- 1) Let me give you another large number, the mass of the Sun: 150000000000 (10 zeros) kg. This can be written as 15×10^{10} kg, according to the powers of 10 notation. We will go a step further and introduce **scientific notation**. In this notation, *all* meaningful numbers (which often *include* zeros that are not sandwiched between nonzero numbers) are written in such a way that the **decimal point is after the first nonzero number**: $15 \times 10^{10} = (1.5 \times 10) \times 10^{10} = 1.5 \times 10^{11}$ kg. Using scientific notation:
 598×10^{22} kg = 5.98×10^{24} kg
 0.0024 m/s = 2.4×10^{-3} m/s.
- 2) Zeros are tricky. Some are meaningful, and some aren't.
 - i) *All leading zeros* are **not** meaningful (**insignificant**), e.g., **0.00234** m. These are placeholder zeros only to indicate the 1's place and 10^{th} , 100^{th} , decimal places, etc. 0.00234 m = 2.34×10^{-3} m (3 significant figures).
 - ii) Trailing zeros **are** meaningful (**significant**) only when **there is a decimal point**, e.g., **100.00** kg. **All zeros are meaningful** here, which is made clear by the presence of the **decimal point**. **The last two zeros mean the measurement was accurate to 100^{th} of a kilogram**. 100.00 kg = 1.0000×10^2 kg (5 significant figures).
 - iii) Trailing zeros are **not** meaningful **without** the decimal point, e.g., **142000** s. These zeros are placeholder zeros only to indicate the 100's, 10's, and 1's place. 142000 s = 1.42×10^5 s (3 significant figures).

- iv) All zeros sandwiched between nonzero numbers are meaningful, e.g., **1004** m = 1.004×10^3 m (4 significant figures).

When using scientific notation, only the meaningful numbers are kept, e.g., 204.200 kg = 2.04200×10^2 kg (6 sig. figs.), 4500 m = 4.5×10^3 m (2 sig. figs.).

Q1) Why are trailing zeros (in bold) significant in the first but not in the second?

- 3) Write numbers using the scientific notations below (Don't forget the units) and state the number of sig. figs.

- i) 340 kg = _____, 2 sf
- ii) 340. kg = 3.40×10^2 kg, 3 sf
- iii) 0.45 m = _____ $\times 10^{-1}$ kg, 2 sf
- iv) 25082 kg = _____
- v) 0.02050 s = _____
- vi) 3600 s = _____
- vii) 123.32 m = _____
- viii) 532.130 kg = _____
- ix) 1000 m = _____ 1 sf
- x) 1000. m = _____
- xi) 1000.0 m = _____ 5 sf
- xii) 1000.000 m = _____
- xiii) 0.00002340920 s = _____ 7 sf
- xiv) 13.450×10 kg = _____
- xv) 299792458 m/s = _____ (EXACT value of the speed of light)
- xvi) 1.000100000 m = _____

- 4) When inputting numbers in scientific notation, **DO NOT** use $^{\wedge}$ or 10^x buttons! Instead, use the EE button, accessed by pressing 2nd then $\frac{\square}{\square}$ buttons. So, to input 1.56×10^{-7} , you first type 1.56 followed by 2nd $\frac{\square}{\square}$, to access EE Then -7. The calculator should display 1.56E-7. The advantage of this button is that scientific numbers in this format don't need to be in parentheses, i.e., no cumbersome typing parentheses. For example, if you want to square 1.56×10^{-7} , press x^2 button after inputting 1.56E-7.

Multiplications and divisions are similarly convenient. For example, $\frac{(1.56 \times 10^{-7})^2 (3.241 \times 10^{23})}{(5.21 \times 10^{-2})^3 (6.35 \times 10^{-7})^{-4}}$ is inputted as 1.56E(-)7 x^2 X 3.241E23 | 5.21E(-)2 $\wedge$ 3 | 6.35E(-)7 $\wedge$ (-) 4 where you should notice the two |'s. That is, anything in the denominator is just followed by |. Also, note, to raise a number to an arbitrary power, you DO need to use the $\wedge$ button. When raising to a negative power, use the negative button next to .,(-), NOT – for subtraction operation. If you do this correctly, you should get $9.067967585 \times 10^{-12}$.

- 5) Perform the following calculations using the EE button. Write answers to 3 sf. Remember, there is no need to enclose numbers in scientific notation in parentheses.

- i) $70(3 \times 10^8)^2 = 70 \times 3E8^2 = 6.3 \times 10^{18}$
(J, if all my mass converted to energy)
- ii) $6.67 \times 10^{-11} \frac{5.98 \times 10^{24}}{(6.37 \times 10^6)^2} =$
(m/s², g_E, acceleration due to Earth's gravity)
- iii) $6.67 \times 10^{-11} \frac{70.0(5.98 \times 10^{24})}{(6.37 \times 10^6)^2} =$
(N, my weight on Earth)
- iv) $6.67 \times 10^{-11} \frac{7.35 \times 10^{22}}{(1.74 \times 10^6)^2} =$
(m/s², g_M, acceleration due to Moon's gravity)
- v) $6.67 \times 10^{-11} \frac{70.0(7.35 \times 10^{22})}{(1.74 \times 10^6)^2} =$
(N, my weight on the Moon)
- vi) $6.67 \times 10^{-11} \frac{(1.99 \times 10^{30})(5.98 \times 10^{24})}{(1.50 \times 10^{11})^2} =$
(N, Gravitational attraction between the Earth and Sun)
- vii) $6.67 \times 10^{-11} \frac{(7.35 \times 10^{22})(5.98 \times 10^{24})}{(3.84 \times 10^8)^2} =$
(N, Gravitational attraction between the Earth and Moon)
- viii) $8.99 \times 10^9 \frac{(1.60 \times 10^{-19})^2}{(5.29 \times 10^{-11})^2} =$
(N, attractive force between the electron and proton in hydrogen)
- ix) $\frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{450 \times 10^{-9}} =$
(J, energy in a single photon of blue light)
- x) $2(1.67 \times 10^{-27})(3.00 \times 10^8)^2 =$
(J, Energy release in proton-antiproton annihilation)